

Geography Ch-18 (Advanced): The Thar Desert - India's Hot Desert & the GIB-Solar Conflict

Tier 2: Advanced | Source: D.R. Khullar + Majid Husain (Geography of India)
India-applied grounded in Qdrant upsc_static book DB + live Current Affairs anchor

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

The Thar (Great Indian) Desert: Origin, Climate, Landforms & Human Adaptation

GS-I (Physical Geography of India) Geography

■ NEWS TRIGGER

Advanced application (D.R. Khullar; Majid Husain, Geography of India). The Thar or Great Indian Desert (western Rajasthan, extending into Gujarat, Punjab, Haryana and Pakistan) is India's only true hot desert - Koppen BWhw - and India's most densely populated desert.

■ INTRO & ORIGIN

The Thar is a Trade-Wind-margin hot desert with a twist: it sits where the SW monsoon's Arabian Sea branch runs PARALLEL to the Aravallis and largely fails to shed rain, aided by descending, dry, divergent air. It shows classic aeolian landforms, xerophytic ecology and remarkable human adaptation.

Aridity causes (Indian context): the Aravallis lie parallel to the Arabian-Sea monsoon current (no orographic barrier to force rain); dry, stable, subsiding air; a former sea-bed (marine origin of some salt flats); and possibly a shifting/lost drainage (the 'Saraswati'). Rain < 25 cm in the west.

■ KEY DATA

Aspect	Detail	Note
Location	Western Rajasthan (+ Gujarat, Punjab)	Koppen BWhw; India's only true hot desert
Rainfall	< 25 cm (west) rising eastward	Highly erratic; summer-concentrated
Landforms	Barchans, seif dunes, playas (dhands)	Aeolian; also inland drainage (Luni)
Vegetation	Khejri, babul, cacti, thorn scrub	Xerophytic; khejri = state tree
Human life	Densest desert population on earth	Pastoralism, IGNP irrigation, tourism

■ STATIC THEORY

- **Why the Thar is arid (Khullar/Majid):** the Aravalli range runs PARALLEL (not perpendicular) to the moisture-bearing Arabian Sea monsoon current, so it fails to force the air to rise and rain; plus descending stable air and a dry continental influence.
- **Landforms:** aeolian - barchans (crescentic dunes), longitudinal (seif) dunes, sand sheets; ephemeral salt lakes/playas (locally 'dhands' e.g. Sambhar - a major salt source); the Luni is the main (inland-draining) river.
- **Ecology:** xerophytic vegetation - khejri (*Prosopis cineraria*, the 'king of the desert', Rajasthan state tree), babul, ber, cacti; fauna adapted to aridity - the Great Indian Bustard, chinkara, desert fox, and the Indian wild ass (in the nearby Rann).
- **Human adaptation:** the Thar is the world's most densely populated desert; traditional water harvesting (tanka, khadin, johad), pastoral nomadism, and modern transformation by the Indira Gandhi Nahar (Rajasthan Canal) bringing irrigated agriculture.
- **Resource frontier:** the Thar now hosts massive SOLAR and wind energy parks (abundant insolation, open land) - central to India's renewable push, but in tension with desert wildlife (see CA).

■ MUST-KNOW FACTS

1. Thar = India's only true hot desert (Koppen BWhw); western Rajasthan; rain < 25 cm in the west.
2. Key aridity cause: Aravallis lie PARALLEL to the Arabian-Sea monsoon current (no rain barrier).

■ UPSC TRAPS

- ✗ The Aravallis block the monsoon and should make the Thar wet.
- ✓ The Aravallis run PARALLEL to the Arabian-Sea monsoon current, so they DON'T force the air up to rain - a key cause of Thar aridity.

3. Landforms: barchan & seif dunes, playas/dhands (Sambhar salt lake); Luni = main inland river.

4. Khejri (*Prosopis cineraria*) = Rajasthan state tree, 'king of the desert'.

5. World's most densely populated desert; transformed by the Indira Gandhi Nahar Project (IGNP).

✗ The Thar is uninhabited like the Sahara's core.

✓ The Thar is the MOST densely populated desert in the world, with pastoralism, canal irrigation and towns.

■ MAINS ANGLE

Explain why the Thar, despite receiving some monsoon winds, remains arid, and assess how the Indira Gandhi Canal and renewable-energy parks are transforming India's desert.

■ STUDY LINK

Geography -> Physical Geography of India -> Thar Desert; links to aeolian landforms (Ch-7) & IGNP

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA Anchor: Great Indian Bustard vs Solar Power Lines - Conservation-Energy Conflict in the Thar

GS-III
(Environment / Conservation) +
GS-II (Judiciary)
Geography -
Current Affairs

■ NEWS TRIGGER

March 2024: the Supreme Court modified its 2021 order that had required ALL overhead power lines in Great Indian Bustard (GIB) habitat to go underground - accepting that blanket undergrounding is impractical for India's clean-energy goals, and setting up an expert committee to balance GIB conservation with solar/wind expansion in the Thar.

■ INTRO & ORIGIN

The Thar's abundant sunshine makes it India's renewable-energy heartland - but the same overhead transmission lines are killing the critically endangered Great Indian Bustard. This is the desert chapter's 'resource frontier vs fragile ecology' theme made concrete.

*GIB (*Ardeotis nigriceps*) - critically endangered, <150 left, mostly in Rajasthan's Desert National Park. Poor frontal vision makes it collide with power lines. 2021 SC order (underground all lines) vs 2024 modification (balanced, expert committee).*

■ KEY DATA

Element	Detail	Significance
Species	Great Indian Bustard (<i>Ardeotis nigriceps</i>)	Critically Endangered; <150 individuals
Threat	Collision with overhead power lines	Poor frontal vision
2021 SC order	Underground lines in GIB habitat	Prioritised conservation
2024 SC modification	Balanced approach + expert committee	Undergrounding impractical for clean energy
Conservation	Project GIB, captive breeding, Desert NP	Bird diverters on lines

■ STATIC THEORY

- **Great Indian Bustard (GIB):** India's most critically endangered large bird; state bird of Rajasthan; flagship of arid-grassland conservation; killed mainly by power-line collisions due to poor frontal vision and heavy body.
- **The dilemma:** the Thar is prime for solar/wind, requiring dense transmission lines - directly conflicting with GIB survival; both clean energy and species conservation are national priorities.

- **Judicial balancing:** the SC's shift from a blanket undergrounding order (2021) to a nuanced, committee-based approach (2024) illustrates the tension between environmental protection and development in environmental governance.

■ MUST-KNOW FACTS

1. Great Indian Bustard (*Ardeotis nigriceps*): Critically Endangered (IUCN); <150 left, mostly Rajasthan.
2. Main threat: collision with overhead power lines (poor frontal vision).
3. 2021 SC order: underground power lines in GIB habitat; 2024: modified to a balanced expert-committee approach.
4. Key habitat: Desert National Park, Rajasthan; conserved via Project GIB & captive breeding.
5. Interim mitigation: bird diverters on existing lines.

■ UPSC TRAPS

- ✗ The GIB is threatened mainly by hunting today.
- ✓ The dominant modern threat is COLLISION WITH POWER LINES in its Thar habitat, alongside habitat loss.
- ✗ The Supreme Court in 2024 ordered all Thar power lines underground.
- ✓ The 2024 order MODIFIED the 2021 blanket-undergrounding directive to a balanced approach via an expert committee.

■ MAINS ANGLE

Discuss the Great Indian Bustard-solar power conflict in the Thar as a case study of balancing renewable-energy expansion with biodiversity conservation.

■ STUDY LINK

Geography -> Thar ecology + GS-III conservation (GIB, renewable energy) & GS-II judiciary; distinct from Thar-greening CA (Ch-7)